Linearity preparation: row overscan
Mathematics and implementation, version 1

Wenu-imaging

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Scope and configuration

This guide describes the preparation kernels and their FITS/recipe/ASDF integration. The kernel passed its actual Mac gate. The new integration requires focused Mac qualification; real corrected ROI products and physical calibration acceptance remain pending. The reviewed design is in the developer design directory:

`linearity-preparation-overscan-asdf-v1.md`.

The detector TOML describes the full detector bands and dimensions. The processing TOML selects the sample columns, trim rectangle, rejection counts, ROIs and preparation controls. The shared Python resolver validates their relationship and freezes both original texts and SHA256 identities. There is no PAF fallback. For CCD65, the full overscan is columns 2097–2144, but the sample is 2100–2142. These distinctions survive in the resolved record.

Exact row correction and arithmetic

For raw frame f, raw row y, and the n enabled scan values sorted in ascending order, the existing trimmed-mean routine returns

$$O_f(y) = \mathrm{pix}\left[\frac{\sum_{k=l+1}^{n-h} v_{f,y,(k)}}{n-l-h}\right]. \tag{1}$$

The sum and division use double precision; pix converts to the native `pixelvalue` type. Low/high rejection is $l = h = 10$ in the selected OmegaCAM profile: 43 reads retain 23, and 44 retain 24. No sigma clipping, smoothing, row fitting, endpoint extrapolation or global-level replacement is introduced. First and last raw rows use their own scan samples. The active-pixel BPM is not applied to electronic scan reads.

The native stages are

$$U_f(x,y) = \mathrm{pix}[D_f(x,y) - O_f(y)], \tag{2}$$

$$S_f(p) = \mathrm{pix}[U_f(p) - B(p)] \quad \text{(flats only)}. \tag{3}$$

The trim uses inclusive FITS coordinates after scan subtraction. An active row j maps to raw row $j + y_{\min} - 1$. Preserve both native roundings: do not merge the subtractions. A supplied master is already scan-corrected and active; it is subtracted once and receives no second scan correction. Bias-role frame preparation has no master subtraction. The existing bias service performs frame admission and stacking; this increment does not change its median or rejection policy. The recipe verifies master configuration, source memberships and lineage before reuse; compatibility is not inferred from pixel values.

Corrected ROI statistics and populations

The paired service first reuses the unchanged raw-pair measurement service. For each configured ROI it uses the common population $\mathcal{P}$ admitted by the declared BPM and finite/raw-saturation checks in both flats and both raw biases. The raw saturation boundary remains 65534 ADU. Pair members retain their original identity, HDU, input index and exact duration.

For corrected member values S_{1p}, S_{2p}, let $N = |\mathcal{P}|$. The services retain ordinary means and sample scatter of each member and their difference:

$$\overline{S}_i = N^{-1} \sum_{p \in \mathcal{P}} S_{ip}, \tag{4}$$

$$s_i^2 = (N-1)^{-1} \sum_{p \in \mathcal{P}} (S_{ip} - \overline{S}_i)^2, \tag{5}$$

$$d_p = \mathrm{pix}[S_{1p} - S_{2p}], \qquad s_d^2 = (N-1)^{-1} \sum_{p \in \mathcal{P}} (d_p - \overline{d})^2. \tag{6}$$

Difference pixel arithmetic follows `image_pair_stats_checked`. The native result stores standard deviations and availability; the native TSV transport stores their squared values and $s_d^2/2$ with explicit definitions. For $N < 2$, sample scatter is unavailable, not zero. A corrected nonfinite value cannot silently remove an otherwise admitted measured pixel: all three corrected population counts must equal the raw count.

Spatial scatter and difference scatter are diagnostic quantities, not assigned independent weights. The shared master cancels algebraically in differences, subject to native rounding. Raw bias-only RON and raw gain diagnostics retain their original definitions. No uncertainty model or physical response is estimated. ROI means remain distinct from the direct physical pixel samples required by the later nonlinear model.

Connection to implementation and verification

`pycavi_linearity_config.resolve` performs TOML resolution only.

`mef_linearity_prepare_frame` validates geometry and finite enabled scan reads, copies the borrowed raw image and uses the existing `mef_overscan_correction_traced` arithmetic. Its optional trace copies each estimate before QC median sorting, preserving physical row order. The legacy interfaces pass a null trace and keep their numerical behavior. The result owns the active image, trace and settings copy.

`mef_linearity_prepare_pairs` corrects the two biases for trace evidence and two flats at a time, reusing raw-pair admission and the existing masked single-image and difference statistics. Results own copied identities, raw diagnostics, corrected ROI statistics and traces. Borrowed inputs stay unchanged; occupied outputs, invalid settings, resource limits and failures do not expose partial completed owners. Custom allocation checks cover the new owners and snapshot owners; they do not establish general Eclipse allocation safety.

Independent ramp fixtures cover both scan widths, raw/active edges, exact singleton-path agreement, common-population analytical moments, and persistent owner-allocation failures. Existing bias-service, sample and raw-pair controls check affected shared paths. Six TOML and five footprint controls check the configuration boundary. Agent Linux success is not actual Mac acceptance or scientific validation. The actual Mac kernel gate passed four native and eleven Python controls; new integration qualification remains separate.

Recipe and immutable ASDF boundary

`pycavi_linearity_recipe.preparation_dependencies` reads the supplied TOMLs and preserved sample/master QC graph. It requires exact scan, trim and stack compatibility with the master snapshot and matches original identities, HDU bindings, accepted/rejected memberships and readout. Legacy header fields travel through a generated native compatibility TOML; they are not scientific configuration overrides. Historical bias QC may have null image digests; the completed bias execution receipt then supplies those fingerprints explicitly.

`prepare_detector` fingerprints input images before native processing, checks them again afterward, and freezes metadata and executable identities. `mefpp_linearity_prepare_command` reads every bias for ordered traces, then streams flat pairs through `mef_linearity_from_files`, using the qualified preparation service and the existing direct-pixel extraction. It reuses the existing master and never rebuilds it. Direct-pixel values are replay checks against the preserved archive, not substitutes for full ROI measurements. No separate singleton extraction campaign is required.

`pycavi_linearity_preparation_qc.publish` uses the unchanged common `QcRecipe`, writer, reader and traversal. Each per-detector product stores corrected individual-exposure means, sample variances, pair differences, original raw diagnostics/populations, ordered bias/flat traces, frozen native transports, both TOML texts and resolved settings, direct-site replay, and the copied source QC graph. Common measurements link to the actual flat inputs. HDU/ROI/exposure identities survive independently of pair iteration order. Original archives remain immutable; existing outputs cannot be overwritten.

`read_product` verifies values against embedded native transports and frozen relative QC links, so relocation does not require original raw images or native output directories. Later model experiments consume this saved boundary without repeating preparation. Absolute noise/covariance stays unavailable and physical fitting stays unperformed. The modelling accessor refuses to imply that missing uncertainty inputs are ready for a fit.

Independent tiny FITS tests cover shuffled HDUs, known row ramps and ROI moments, all ten bias memberships, original direct-site values, duplicate and resource refusals, source preservation, ASDF roundtrip and relocation without native files. Focused file-adapter allocation controls and the unchanged singleton command check shared interfaces. None is scientific acceptance of observational calibration or a general dependency memory-safety claim.